

Extending NLP heuristics to Speaker recognition using Discrete Speech Units

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1. INTRODUCTION

The growing success of self-supervised models in speech processing aroused interest in discrete speech units (DSUs) derived from contextualized representations like HuBERT and Wav2Vec. These units have demonstrated effectiveness in content-based tasks, such as automatic speech recognition (ASR) and expressive speech generation [1, 2]. However, their application to speaker recognition, a task fundamentally reliant on low-level acoustic features like phonetics, remains under-explored. This work investigates the potential of DSUs for capturing speaker-specific information.

This study aims to determine if DSUs can effectively identify speakers. A key question is whether this identification is driven primarily by the linguistic content of the speech, or if DSUs also capture and utilize phonetic cues related to the speaker's voice. Two publicly available datasets were utilized: LibriSpeech [3] and VCTK [4]. LibriSpeech, a large collection of read speech, offers the advantage of diverse textual content for each speaker, while VCTK provides recordings of multiple speakers reading the same text, enabling a focused analysis of vocal characteristics.

K-means clustering was applied to contextualized representations from a pre-trained self-supervised model (HuBERT-Large) to derive DSUs. This process yielded discrete cluster indices, which were treated as pseudo-text. Next, standard text processing techniques, such as TF-IDF and CountVectorizer were used to generate vector representations of these DSU sequences. Then several classification methods were explored, including Logistic Regression, Multinomial Naive Bayes, Decision Tree Classifier, Random Forest, and SVM. Also, the DSUs were represented as embeddings, and a simple recurrent neural network (RNN) was used for classification.

To establish a content-based speaker recognition baseline, a transcript-based approach is implemented for both LibriSpeech and VCTK datasets. This baseline, expected to be limited by its reliance solely on linguistic content, serves as one of the baselines for evaluating the contribution of DSUs. Besides this, we also trained an RNN on the same features used for extracting DSUs, which serves as a topline for our experiments.

This work contributes to the broader research on speaker recognition by exploring the novel application of DSUs with Count and TF-IDF vectorizer. By analyzing the performance of DSUs in comparison to both text-based and phonetic baselines, this study aims to clarify the nature of speaker-specific information captured by these discrete units. The findings presented here have implications for various applications, including speaker separation, voice identification, and forensic speaker identification.

2. DATASETS

Two datasets were utilized for this project. The first dataset, LibriSpeech [3], consists of speech extractions from English audiobooks, where each speaker reads a unique text. In contrast, the VCTK dataset [4] contains recordings in which multiple speakers read the same text which is suitable for the phonetic analysis. For both datasets, 30 speakers have been selected with the highest number of recorded samples to ensure a balanced sample and reduce the computational time.

From each dataset, we extracted the following columns:

- **file**: the path to the downloaded audio file
- **text**: the transcription of the audio file
- **speaker_id**: unique id of the speaker
- **DSU**: compact representations for speech signals that were obtained by k-means clustering of encoded audio (for more details see section 3.1)
- **DSU_Deduplicated**: DSUs representation after data preprocessing (for more details see section 3.1)

Each dataset was split into 88% for the training set and 6% each for the development and test sets.

LibriSpeech contains a total of 5,620 records, with each speaker having an average of 335 records (see the distribution in Figure 1).

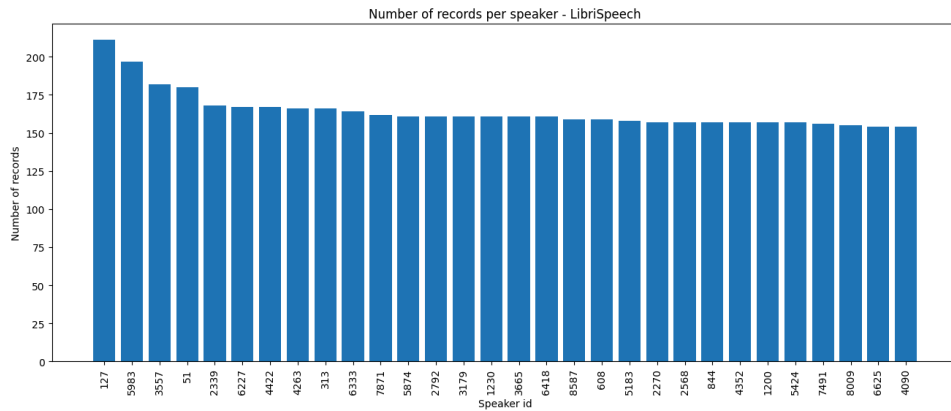


Figure 1. Distribution of number of records per speaker in LibriSpeech dataset.

VCTK consists of approximately 11,200 records, with each speaker having an average of 164 records (see the distribution in Figure 2).

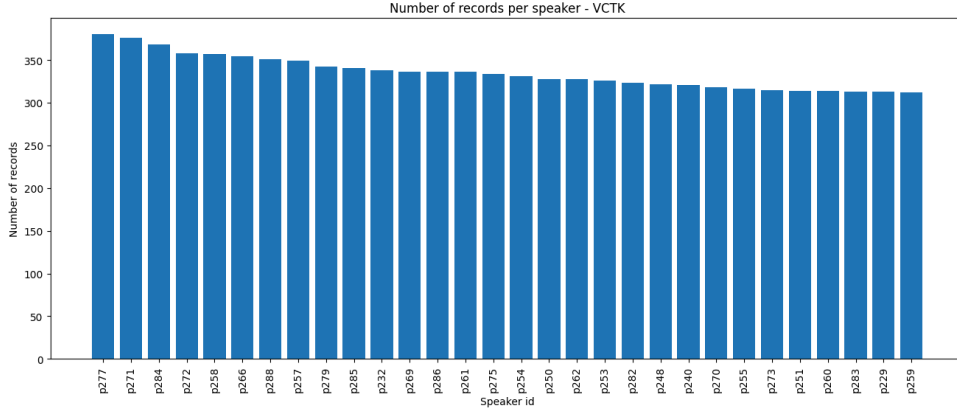


Figure 2. Distribution of number of records per speaker in VCTK dataset.

Each text is repeated an average of four times across different speakers. However, some texts appear more than 20 times, with a maximum repetition of 29 occurrences. (the distribution is shown in Figure 3).

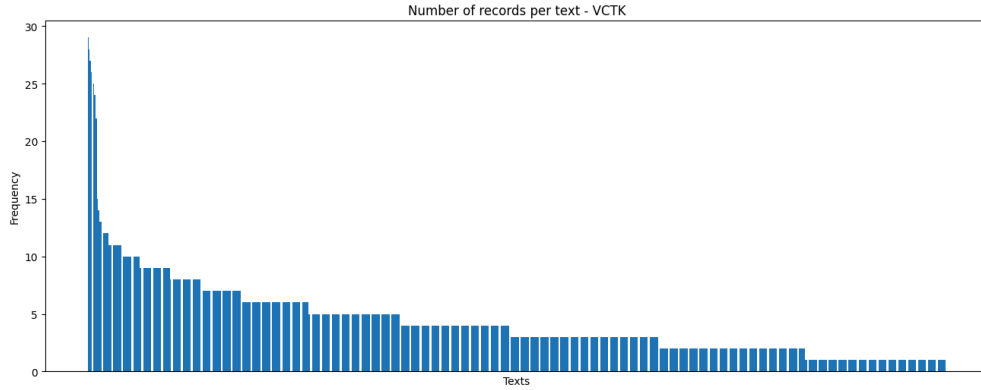


Figure 3. Distribution of number of records per text in VCTK dataset.

3. METHODS

3.1. Data Preprocessing

For DSU extraction, we apply an audio encoder followed by K-means clustering.

Audio encoder: To extract speech features, we utilize HuBERT-Large, a model pre-trained on 60k hours of speech data from the LibriVox dataset. This model is not fine-tuned; instead, we use it offline to obtain continuous features. These features are then used to train a K-means model, which generates the DSUs.

K-means: Initially, we train a K-means clustering model on the continuous speech features extracted by the HuBERT encoder. The embeddings from the 22nd layer of the HuBERT model are then partitioned into $K = 2000$ clusters, a choice based on prior research. To avoid test set information leakage during the training of the K-means model, we ensure that it is trained only on features from the training set. Once the K-means model is trained, the original continuous vectors are replaced with their corresponding cluster IDs, transforming the continuous speech representations into a discrete set of speech units (DSUs).

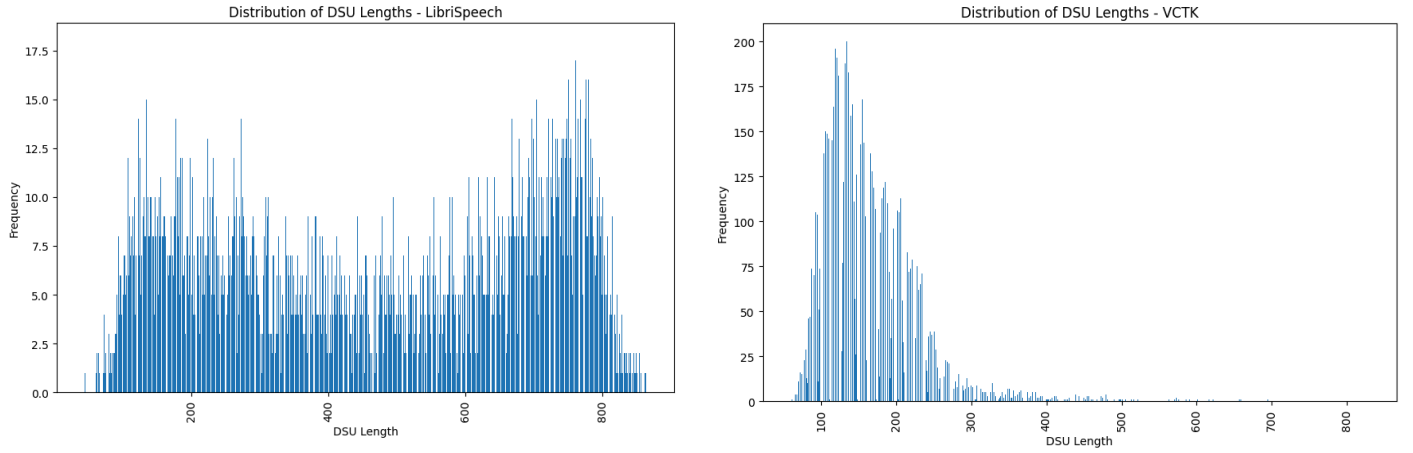


Figure 4. The distributions of DSU lengths for LibriSpeech (left) and VCTK (right) datasets.

Figure 4 presents the distribution of DSU lengths among all records, with an average length of 466 units for LibriSpeech dataset and 165 units for VCTK. While VCTK has a bell-shaped left-skewed distribution, LibriSpeech has multiple peaks and frequency values vary which implies that DSU lengths are diverse.

DSU Deduplication: As an additional preprocessing step, we apply deduplication by removing consecutive duplicate tokens. This step is important because repetitive DSUs can be redundant, and eliminating them reduces unnecessary information while preserving the essence of the speech content. By compacting the representation, deduplication also helps improve efficiency in downstream tasks. After deduplication, the average DSU length is reduced to 335 units for the LibriSpeech dataset and 115 units for VCTK. The length distribution is shown in Figure 5. The distributions follow the same tendency as for the DSUs.

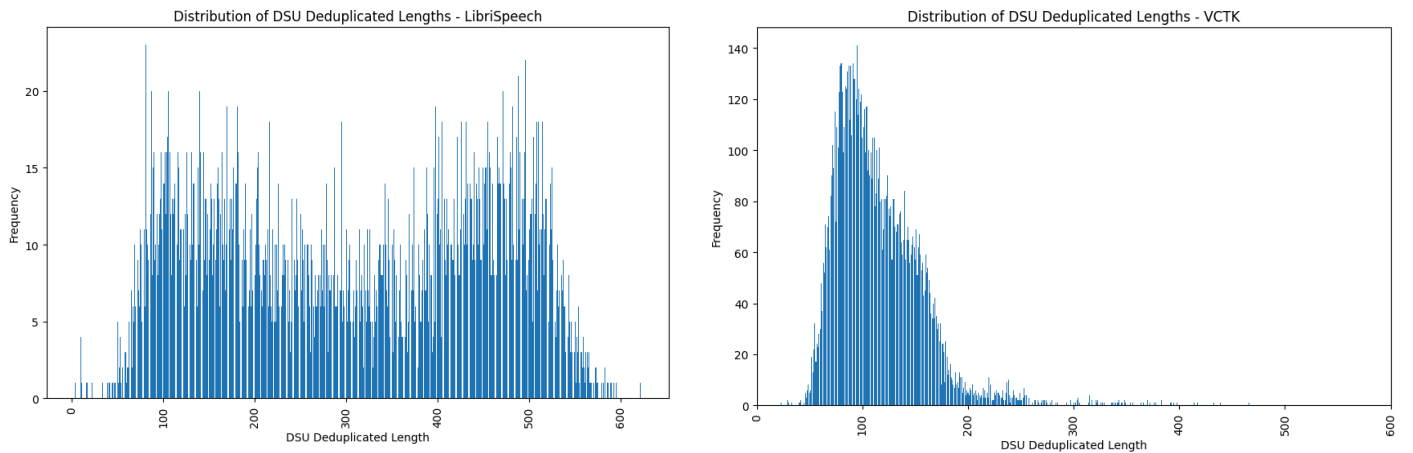


Figure 5. The distributions of deduplicated DSU lengths for LibriSpeech (left) and VCTK (right) datasets.

For analysis, both the original DSU sequences and the deduplicated DSUs will be used for comparison to evaluate their effectiveness in capturing speaker-specific

information.

3.2. Feature Engineering

Since DSUs function similarly to subwords in NLP, they can be structured for classification using NLP techniques such as CountVectorizer and TF-IDF, incorporating both unigrams and bigrams to capture statistical patterns. We hypothesize that because DSUs are derived by clustering high-dimensional contextual vectors from a speech model, their variations should retain phonetic information. This information can be effectively highlighted using NLP methods like CV and TF-IDF, which were originally designed for text but operate in a similar context.

3.3. Experimental setup

To systematically evaluate the applicability of CV and TF-IDF in the context of Discrete Speech Units (DSUs), we conducted a series of experiments using different input representations. First, we performed classification based on speech transcripts to establish a baseline using textual information. With this experiment, we seek to establish that speakers cannot be simply classified based on the content of their speech. Next, we explored classification using statistical methods applied directly to DSUs (and deduplicated DSUs), leveraging their structured representation. Additionally, we trained a topline RNN-based classification model using continuous speech features to assess how well raw audio representations perform. Finally, we treated DSUs as text tokens, applying LSTM to analyze their effectiveness in capturing meaningful patterns. These experiments provide a comprehensive analysis of how DSUs compare to other modalities in classification tasks.

After fitting CV and TF-IDF on DSUs and text, we experimented with several classification models to find the best classification setting. In the following section, we describe all the classification models we used for Vectorizers (CV and TF-IDF) for DSUs and text. Following it, we describe the architecture of the RNN and LSTM model used with continuous features and DSUs respectively.

3.4. Models

RNN architecture for continuous features: Since we use the same features that were used to train the k-means model, initially, we have a sequence of 1024 dimensional contextual vectors. We feed this sequence directly to an RNN with 3 layers and a hidden state dimension 256. Also since we have the entire audio apriori, instead of using the classical unidirectional RNN, we use its bidirectional counterpart. For classification, the hidden state from the last time step (concatenated from both directions) is passed through a linear layer which projects it to a 30-dimensional vector used for classification after applying softmax.

LSTM architecture for DSUs: The input consists of sequences mapped to a 128-dimensional embedding space using an embedding layer that supports 2000 DSU tokens

(+1 padding token), with a designated padding index of 2000. These embeddings are then processed by a 4-layer bidirectional LSTM, where each direction has a 128-dimensional hidden state, resulting in a 256-dimensional combined representation at each time step. A dropout layer ($p=0.2$) is applied to prevent overfitting. For classification, the final hidden state is passed through a fully connected classifier consisting of four linear layers with progressively decreasing dimensions: $256 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 30$. Each intermediate layer uses a Tanh activation function to introduce non-linearity, while the final layer outputs a 30-dimensional vector, which is subsequently processed using softmax for classification.

Classification models used with Vectorizers: We explored traditional classification methods such as Logistic Regression, Multinomial Naive Bayes, Decision Tree Classifier, Random Forest, and SVM. Based on the results, the best-performing model is selected, followed by hyperparameter tuning using Grid Search.

4. RESULTS

Several approaches were applied to analyze DSUs. Initially, DSU sequences were treated as textual data and transformed into vector representations using statistical text processing techniques. Term frequency was examined in its raw Counter form and inverse document frequency (TF-IDF) weighting. Vocabulary construction included both unigrams and bigrams.

The generated feature representations were evaluated using vanilla classification models without hyperparameter tuning to determine the most suitable classifier. Model performance was assessed based on accuracy, precision, recall, and F1-score. Given the time constraints of the experiments, the final model which is Multinomial Naive Bayes was selected not only for its high-performance metrics but also for its training efficiency. The results of the vanilla model selection are shown in Figures 6, 7.

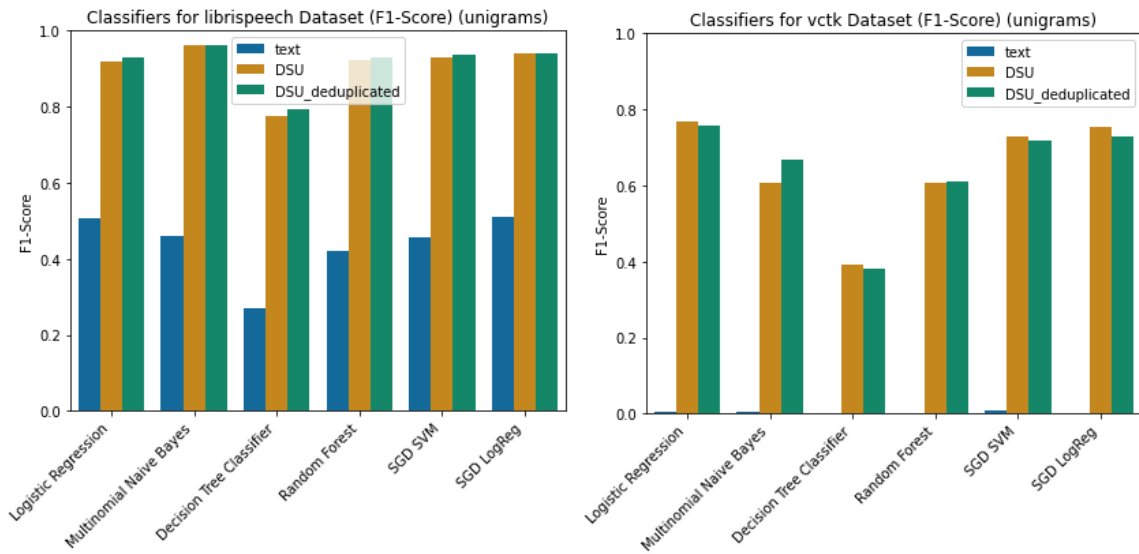


Figure 6. Vanilla models prediction quality for the CounterVectorizer approach, using unigrams vocabulary.

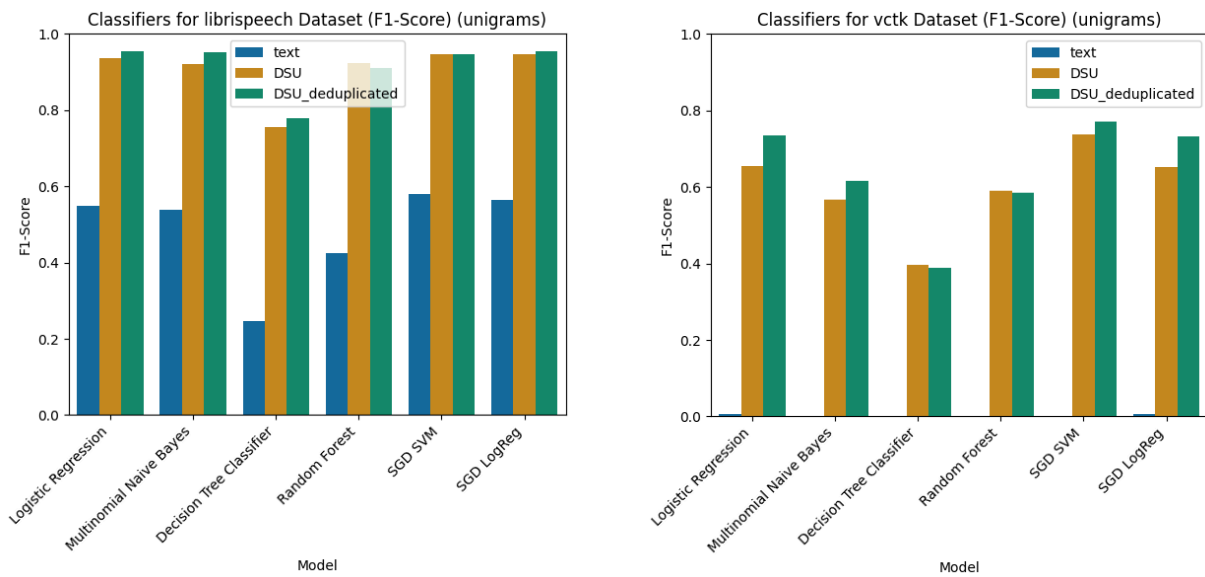


Figure 7. Vanilla models prediction quality for the TF-IDF approach, using unigrams vocabulary.

The more descriptive results can be found in the following notebooks: Vectorizers/CV.ipynb and Vectorizers/Tfidf.ipynb with the results for the CountVectorizer and Tf-Idf respectively.

Hyperparameter tuning was conducted using grid search with 5-fold cross-validation for each dataset and its corresponding feature representations. After optimization, the final model was tested using the best-performing hyperparameters, and confusion matrices were

generated to further analyze classification performance. The results of the quality of model predictions after selecting hyperparameters are presented in Table 1.

Table 1. Quality of models predictions on the test dataset (LSTM model, else RNN)**

Features	F1-score				
	CounterVecorizer		TF-IDF		RNN based
	MNB, unigrams	MNB, bigrams	MNB, unigrams	MNB, bigrams	
LibriSpeech					
DSUs	0.9624	0.9800	0.9575	0.9800	–
DSUs Deduplicated	0.9681	0.9828	0.9594	0.9772	0.7348**
Transcript (text)	0.5431	0.4090	0.5615	0.4480	–
Continuous	–	–	–	–	0.9403
VCTK					
DSUs	0.6125	0.6706	0.5667	0.6599	–
DSUs Deduplicated	0.6679	0.7071	0.6352	0.6934	0.3677**
Continuous	–	–	–	–	0.8939

Overall, DSUs with vectorizers perform exceptionally well on Librispeech, with F1-scores exceeding 0.96, particularly when using bigrams (0.98). Deduplicating DSUs slightly improves performance, indicating that removing consecutive duplicates enhances speaker distinction. However, the difference is only in the 3rd character after the dot. In contrast, text transcripts perform significantly worse (F1-scores between 0.409 and 0.561), meaning that speaker identity is not well captured by text.

On the VCTK dataset, performance is generally lower. DSUs achieve F1-scores between 0.6125 and 0.6706, but deduplication provides improvements. Such low compared to LibriSpeech results mean that both text and phonetic features of a speaker are important for the speaker identification.

The RNN model performed well with continuous features, requiring minimal tuning to achieve good results. However, when working with DSUs, we observed significant overfitting on both the LibriSpeech and VCTK datasets using a vanilla RNN. Despite experimenting with dropout regularization, batch normalization, and reducing model complexity, the vanilla RNN consistently failed to generalize. To address this, we switched to an LSTM, which significantly reduced overfitting on the LibriSpeech dataset. However, even

with the LSTM, overfitting persisted on the VCTK dataset. The training curves for both datasets are shown below.

5. DISCUSSION

Text features are not enough, phonetics are important: Our results indicate that while text-based features provide some level of speaker discriminability, phonetic information plays a crucial role in capturing speaker identity. In LibriSpeech, where speakers read different texts, the classification performance using CV/TF-IDF vectorized transcripts is significantly higher than a random baseline, suggesting that certain statistical patterns in word choice and phrasing may differentiate speakers. However, in VCTK, where all speakers read the same content, the performance of text-based features diminishes, reinforcing the idea that text alone is insufficient for speaker classification in such settings. This highlights the importance of phonetic cues, as different speakers articulate the same text with unique acoustic characteristics that are not captured by text-based methods.

Continuous features outperform DSU-based methods: In our experiments, models trained on continuous speech features consistently outperformed those relying on discrete speech units (DSUs). This suggests that fine-grained acoustic details present in continuous representations provide a richer signal for speaker classification. Interestingly, in LibriSpeech, vectorizers (CV/TF-IDF) performed slightly better than continuous features, but we hypothesize that with proper hyperparameter tuning of the RNN trained on continuous features, this gap could be bridged. Due to time constraints, we did not fully optimize the RNN architecture, which might have limited its potential performance. On the other hand, in VCTK, where phonetic variations are crucial for distinguishing speakers, continuous features clearly outperformed DSU-based methods, reinforcing the advantage of preserving fine-grained acoustic information over discretized speech representations.

DSUs are prone to overfitting in limited data regime: While experimenting with approximately 150 training datapoints in LibriSpeech (300 in VCTK), we found that vanilla RNNs consistently failed to generalize. This could be attributed to the model's inability to capture long-range dependencies or complex patterns, prompting us to switch to an LSTM. Switching to LSTM greatly alleviated the overfitting issue for LibriSpeech. However, for VCTK, the gap between training and validation performance remains, suggesting that further improvements are needed. A plausible explanation for this poor performance on VCTK, despite using LSTM, is that most speakers read the same sentence, causing the model to encounter DSUs from entirely different sentences during testing. This issue is amplified in the low-data regime, where the training set lacks sufficient sentence diversity. The problem may stem from how the embedding layer of the LSTM is trained. In contrast, for continuous features, where embeddings are derived from a highly rich and contextualized HuBERT model, even a simple RNN performs well in this setting.

6. CONCLUSION

In this study, we investigated the effectiveness of Discrete Speech Units (DSUs) in speaker classification and compared them with traditional text-based features and continuous speech representations. Our results demonstrate that while text-based features provide some speaker discriminability, phonetic information, captured through continuous speech features, plays a crucial role in accurately identifying speakers, particularly in datasets like VCTK. DSU-based methods, while effective in Librispeech, faced challenges in the limited data regime, exhibiting significant overfitting, which was addressed to some extent by switching to LSTM models. However, even with LSTM, the challenge of generalizing on the VCTK dataset remained, likely due to the lack of sentence diversity in the training set.

The comparison of DSU vectorization with text transcripts highlighted the importance of phonetic features in speaker classification, as text alone was found to be insufficient. Continuous features, despite requiring minimal tuning, outperformed DSU-based methods, particularly in scenarios where phonetic variations play a key role. Additionally, we believe that more data could have significantly improved the performance of RNN-based models, especially in overcoming the overfitting observed in the limited data settings. Overall, this work underscores the need for more refined models and better utilization of phonetic features in speaker identification tasks, particularly in low-data settings like VCTK, and opens the door for further exploration of hybrid approaches that combine both phonetic and text-based information.

7. REFERENCES

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